

Markscheme

Mathematics: analysis and approaches

Practice question bank

10. (a)

Consider the expression $\frac{\sqrt[3]{x^4} \times x^{-\frac{1}{2}}}{\sqrt{x^{-3}}}$.

Simplify it to the form x^k , and find k .

$$\sqrt[3]{x^4} = x^{\frac{4}{3}} \text{ and } \sqrt{x^{-3}} = x^{-\frac{3}{2}} \quad \text{A1}$$

$$\text{combine the numerator: } x^{\frac{4}{3} - \frac{1}{2}} = x^{\frac{5}{6}} \quad (M1)$$

$$\text{divide by } x^{-\frac{3}{2}} \quad (M1)$$

$$x^{\frac{5}{6} + \frac{9}{6}} = x^{\frac{14}{6}} \quad \text{A1}$$

$$k = \frac{7}{3} \quad \text{A1}$$

[5 marks]

(b)

Hence evaluate the original expression when $x = 8$.

$$8^{\frac{7}{3}} = (\sqrt[3]{8})^7 = 2^7 \quad (M1)$$

$$= 128 \quad \text{A1}$$

[2 marks]